In [32]:

```
import pandas as pd
import numpy as np
import time
import re
from selenium import webdriver as wd
from bs4 import BeautifulSoup
from datetime import datetime
```

In [33]:

```
url1 = 'https://www.youtube.com/playlist?list=PLVUMrRRWz02pSbuW-RhxroVI0xhgycZvb'
url2 = 'https://www.youtube.com/playlist?list=PLVUMrRRWz02qsQn3dPX67df-yVIpJ8o3Z'
```

text crawling

- 우리학교 유튜브에서 2021학년도 1학기와 2학기 DS Interface에 올라온 논문들의 논문제목을 크롤링해서 아래의 예시와 같이 출력해주시면 됩니다
- 동영상 앞에 [DS Interface]라고 표기된 부분은 제외하고 제목만 출력

1학기 논문 제목

In [ ]:

```
driver = wd.Chrome(r"D:₩9. 대학원 수업₩21_2학기₩비정형데이터₩과제7₩chromedriver.exe")
url = (url1)
driver.get(url)
```

In [35]:

```
soup = BeautifulSoup(driver.page_source, 'html.parser')
```

In [36]:

```
video_name = soup.select('a#video-title')
```

In [ ]:

```
titles = []
for i in range(len(video_name)) :
    titles.append(video_name[i].text.strip().replace('[DS Interface] ', ''))
#print(titles)
```

In [38]:

```
df = pd.DataFrame({"2021학년도 1학기" :titles})
df
```

Out[38]:

	2021학년도 1학기
0	Generalizing from a Few Examples: A Survey on ...
1	Categorization of YouTube Videos by Video Samp...
2	Predicting Named Entity Location using Twitter
3	Reformer: The Efficient Transformer
4	Estimating Training Data Influence by Tracing ...
5	GloVe: Global Vectors for Word Representation
6	CutMix: Regularization Strategy to Train Stron...

	2021학년도 1학기
7	[DS interface] Link prediction based on graph ...
8	A recommender system for tourism industry usin...
9	Delving Deep into Rectifiers: Surpassing Human...
10	A new similarity measure for collaborative fil...
11	The Lottery Ticket Hypothesis: Finding Sparse,...
12	A Closer Look at Few Shot Classification
13	Technology opportunity discovery under the dyn...
14	Effective Korean sentiment labeling method usi...
15	Graph Convolutional Networks(GCN): a Review
16	Predicting Popularity of Online Videos Using S...
17	VoxelNet: End-to-End Learning for Point Cloud ...
18	Grad CAM: Visual Explanations from Deep Networ...
19	Aligning AI with Shared Human Values
20	ITEM2VEC: NEURAL ITEM EMBEDDING FOR COLLABORAT...
21	A deep learning methodology for automatic extr...
22	Over-Sampling Methods for Imbalanced Dataset
23	Automatic Database Management System Tuning Th...
24	Learning loss for Active learning

2학기 논문 제목

In [39]:
```
url = (url2)
driver.get(url)
```

In [40]:
```
soup = BeautifulSoup(driver.page_source, 'html.parser')
```

In [41]:
```
video_name = soup.select('a#video-title')
```

In [42]:
```
titles = []
for i in range(len(video_name)) :
    titles.append(video_name[i].text.strip().replace('[DS Interface] ', ''))
#print(titles)
```

In [43]:
```
df = pd.DataFrame({"2021학년도 2학기" :titles})
df
```

Out [43] :

	2021학년도 2학기
0	Noise-Filtering Methods for Noisy Robust Adaboost
1	Interactive Visualization with Dash, Plotly an...
2	The use of machine learning algorithms in reco...
3	Undefined Class-label Detection vs Out-of-Dist...
4	FixMatch: Simply SSL with Consistency and Conf...
5	Principles and Practice of Explainable Machine...
6	You only look once: Unified, real-time object ...
7	Forecasting emerging technologies: A supervise...
8	A hybrid big data analytical approach for anal...
9	A Closer Look at Accuracy vs. Robustness
10	MAD-GAN: Multivariate Anomaly Detection for Ti...
11	An Image is Worth 16x16 Words: Transformers fo...
12	BertGCN : Transductive Text Classification by ...
13	Convolutional Neural Networks for Sentence Cla...
14	Difficulty-aware Meta-learning for Rare Diseas...
15	Neural Architecture Search with Reinforcement ...
16	Class balanced loss based on effective number ...
17	Homomorphic Encryption : Key of Privacy Preser...
18	Forecasting Crypto-Asset Price Using Influence...
19	Knowledge distillation : A good teacher is pat...
20	RNN-SURV: A Deep Recurrent Model for Survival ...
21	Session-based Recommendation with Graph Neural...
22	World Model
23	Universal Sentence Encoder
24	A novelty detection patent mining approach for...
25	Stop explaining black box machine learning mod...
26	Uncovering the dynamics of market convergence ...
27	Presto: SQL on Everything
28	LSTM-based Encoder-Decoder for Multi-sensor An...
29	Exploring the Limits of Out of Distribution De...